

can also collect data on human-object interactions to train machine learning models for grasping, manipulation, and autonomous decision-making.

The lightweight MediaPipe framework allows the system to run efficiently on edge platforms such as Raspberry Pi, Nvidia Jetson, and standard PCs. Fig. 1 shows the author's prototype robot tracking a face in real time while detecting facial landmark points. Components required to build the system are listed in the Bill of Materials table.

Robot design and assembly: Robot face

In this design, a Raspberry Pi 4 with 2GB RAM serves as the main controller. Nvidia Jetson and Orin boards can also be connected and are recommended for improved performance. The robot's face is based on the open source InMoov robot design. Most components are 3D-printed, along with custom mounts for the camera and Raspberry Pi. Additional single-board controllers are installed inside the head to control servos and internal mechanisms.

First, assemble the 3D-printed parts of the robot's head. Then install the servos to enable eye and head movement. Refer to the assembly instructions here:

<https://inmoov.fr/eye-mechanism>

Refer to Fig. 2 for guidance. After this step, proceed with assembling the eyes.

Adding camera. Place the camera inside the 3D-printed eye housing so that the robot can capture visual input. Fix the camera on one side and assemble the remaining eye components. Finally, mount the assembled eyes onto the servo-driven mechanism in the robot's head. Refer to Fig. 3.

Adding computing. Next, install the robot's computing unit, which performs processing tasks, runs AI-based image processing, and

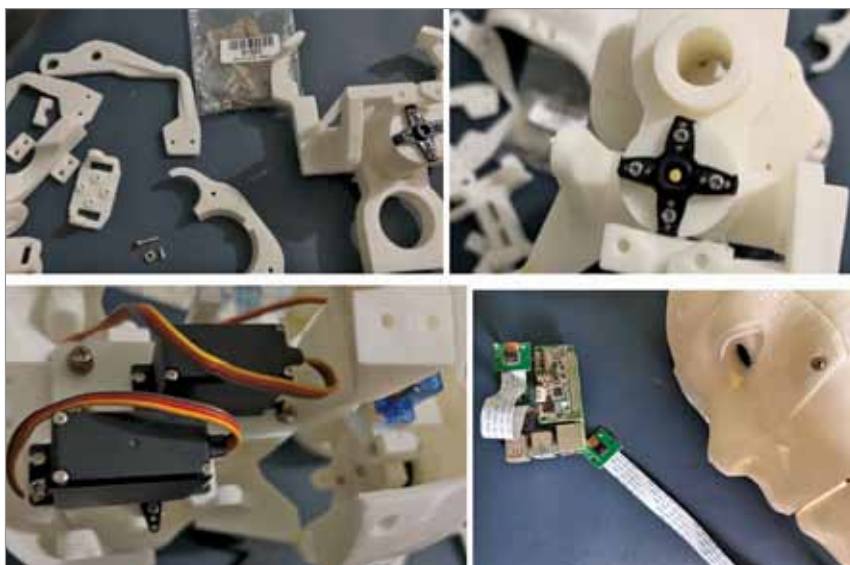


Fig. 2: Robot head assembly and servo motor installation



Fig. 3: Camera mounted inside the robot's eye

supports machine learning models. Fig. 4 shows this setup.

A Raspberry Pi is used due to its compact size, allowing it to fit inside the robot's head. Mount it at the top inside the head and cover it with the remaining 3D-printed parts. Before closing the head, connect the power

supply and servo motors. Refer to Fig. 5 for the circuit diagram.

Circuit diagram

Fig. 5 shows the circuit diagram built around a Raspberry Pi 4 with two servo motors. A Voice AI HAT with inbuilt pins A, B, C, and D is